

Trader Performance vs. Bitcoin Market Sentiment

An Exploratory Analysis of Hyperliquid Trading Data and the Fear & Greed Index

Data Science Assignment — Round 0 Submission

Executive Summary

This analysis joins 211,224 Hyperliquid trade records from 32 accounts against the daily Bitcoin Fear & Greed Index to test whether market-wide sentiment is associated with trader profitability, positioning, activity, and risk. Across every metric examined, sentiment regime was meaningfully associated with trading outcomes, and traders in this dataset behaved contrarian to the sentiment label itself — leaning long during Fear and short during Greed, a posture that coincided with stronger results in the euphoric regimes.

The strongest finding is that Extreme Greed produced the best risk-adjusted returns (highest average PnL, lowest volatility) while capital was, if anything, sized more conservatively in that regime and more aggressively during Fear — the regime with weaker average outcomes. This mismatch between conviction (position sizing) and historical edge is the clearest actionable signal in the data.

1. Datasets & Methodology

1.1 Data Sources

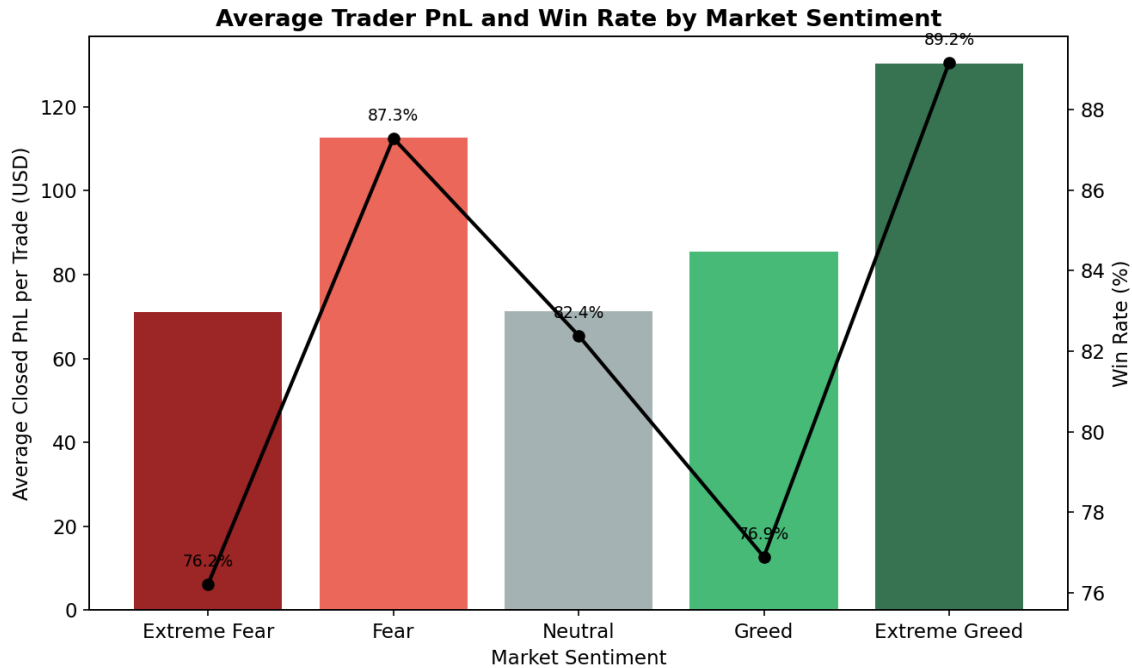
- **Hyperliquid Historical Trade Data** — 211,224 rows covering 32 accounts and 246 symbols, December 2024 to May 2025. Each row is a fill/event with execution price, size, side, direction, closed PnL, fees, and a timestamp.
- **Bitcoin Fear & Greed Index** — 2,644 daily records from February 2018 to May 2025, with a 0–100 index value mapped to five labels: Extreme Fear, Fear, Neutral, Greed, Extreme Greed.

1.2 Approach

- Trade timestamps were converted to calendar dates and joined to the sentiment index on date (99.997% match rate; 6 of 211,224 rows fell outside the index's coverage and were dropped).
- Profitability metrics use only trades with non-zero Closed PnL — these represent realized, closing trades, since position-opening events carry PnL = 0 by construction.
- Directional bias metrics use only 'Open Long' / 'Open Short' events, isolating new-position decisions from closes and other event types.
- All findings are descriptive/exploratory (correlational, not causal) and are based on a comparatively short trade-history window relative to the multi-year sentiment index.

2. Profitability by Market Sentiment

Average realized PnL per trade and win rate both scale with how greedy the market is, peaking in Extreme Greed and bottoming out in Extreme Fear and Neutral.

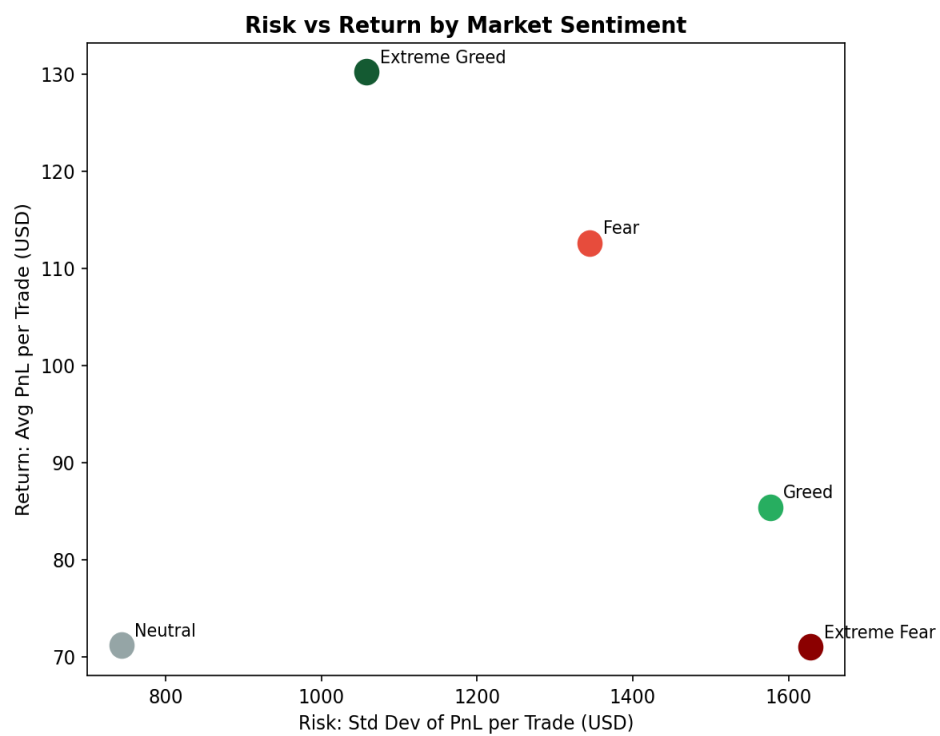


Sentiment	Trades	Avg PnL/Trade (\$)	Win Rate	Total PnL (\$)
Extreme Fear	10,406	71.03	76.2%	739,110
Fear	29,808	112.63	87.3%	3,357,155
Neutral	18,159	71.20	82.4%	1,292,921
Greed	25,176	85.40	76.9%	2,150,129
Extreme Greed	20,853	130.21	89.2%	2,715,171

Extreme Greed stands out on every profitability metric: the highest average PnL per trade (~\$130) and the highest win rate (~89%). Fear also shows a strong win rate (87%) despite a lower average PnL, suggesting many small wins rather than few large ones. Extreme Fear and Neutral are the weakest regimes on both dimensions.

3. Risk vs. Return by Sentiment

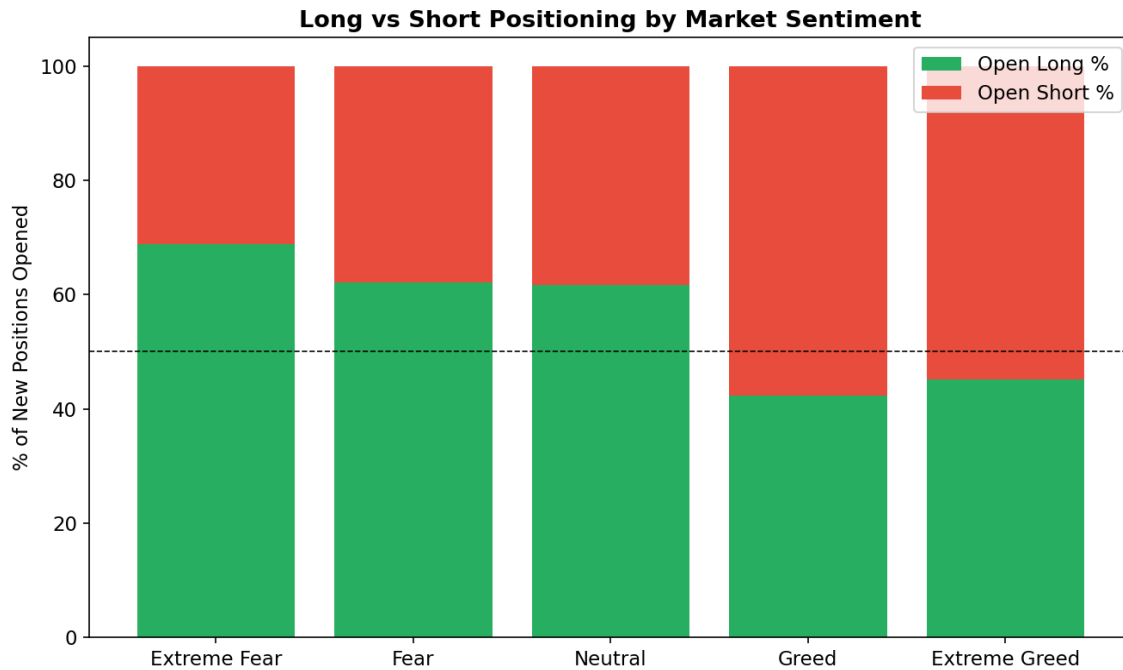
Plotting average PnL (return) against its standard deviation (risk) per sentiment class reveals which regimes offer the best risk-adjusted trade-offs.



Extreme Greed sits in the most favorable quadrant: highest average return combined with the lowest volatility of the five regimes. Fear and Greed themselves are the most volatile — trading outcomes are least predictable when sentiment is directional but not yet extreme.

4. Positioning Behavior: Long vs. Short Bias

Isolating new-position events (Open Long / Open Short) shows a clear and consistent contrarian pattern relative to the sentiment label.



Sentiment	Open Long %	Open Short %
Extreme Fear	68.8%	31.2%
Fear	62.1%	37.9%
Neutral	61.7%	38.3%
Greed	42.3%	57.7%
Extreme Greed	45.1%	54.9%

Traders open predominantly long positions when the market is fearful (up to 69% long in Extreme Fear) and flip to majority-short as sentiment turns greedy (up to 58% short in Greed). This is a textbook contrarian posture — fading the crowd's emotional extreme — and it lines up with the stronger win rates observed in the greed regimes, where shorting into apparent euphoria was rewarded in this sample.

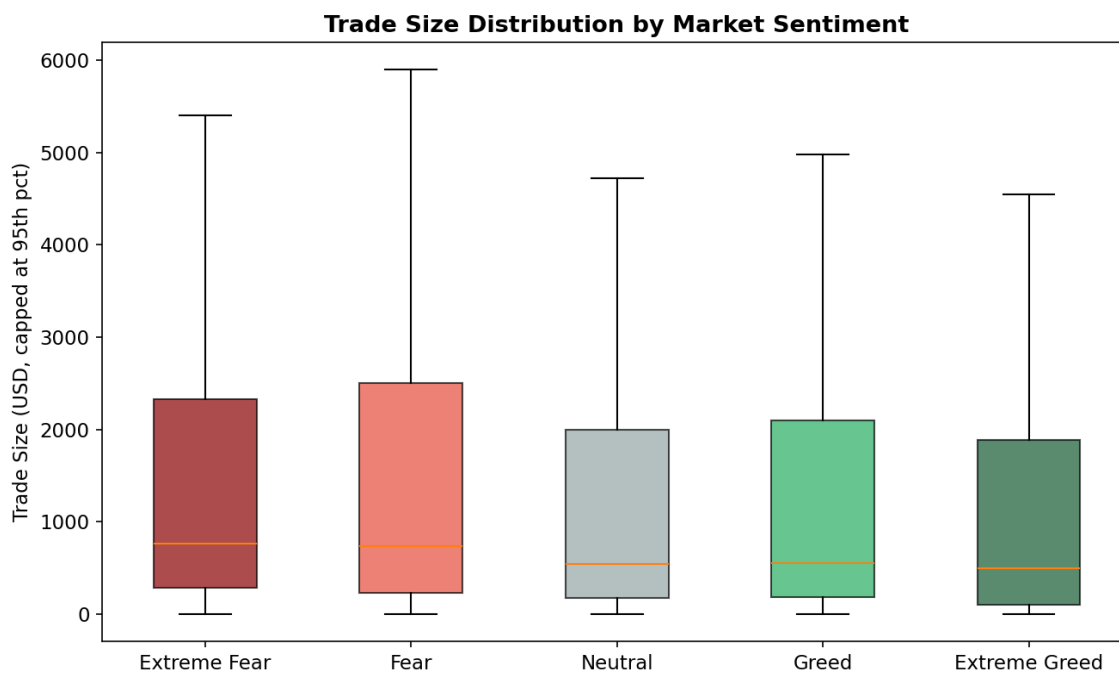
5. Trading Activity & Sentiment Over Time

Daily trade counts were compared against the daily Fear & Greed Index value across the full trading window.



- **Trade count vs. sentiment value** (Pearson $r \approx -0.24$): traders in this dataset transact more when the market is anxious, and quiet down as greed builds.
- **Total daily PnL vs. sentiment value** ($r \approx -0.08$): the level of the index alone is a weak predictor of aggregate daily PnL — positioning and trade selection matter more than the sentiment reading itself.

6. Trade Sizing by Sentiment



Average trade size is largest during Fear (~\$7,816) and smallest during Extreme Greed (~\$3,112) — the inverse of where the strongest average outcomes were observed. Traders are committing more capital per trade in the regime that historically produced weaker average PnL, and less capital in the regime with the best risk-adjusted returns.

7. Trader Concentration & Consistency

Performance is far from uniform across the 32 accounts in the dataset. The top 10% of accounts (3 traders) generated approximately 46% of total realized PnL, while a handful of accounts posted large net losses (as low as -\$167,621 for the weakest account).

Per-account breakdowns by sentiment also show meaningful heterogeneity: some traders are consistently profitable across every regime, while others are strongly regime-dependent — performing far better in Fear/Greed than in Neutral markets, or vice versa. This suggests individual trader skill/style is a stronger driver of outcomes than sentiment alone, and is a natural candidate for a follow-up clustering or segmentation analysis.

8. Key Takeaways

- **Sentiment regime predicts profitability.** Average PnL per trade and win rate both rise with sentiment, peaking in Extreme Greed (~\$130/trade, ~89% win rate) and troughing in Extreme Fear/Neutral (~\$71/trade).
- **Positioning is contrarian to the index label — and that's where the edge was.** Traders lean long in Fear (up to 69%) and short in Greed (up to 58%) — fading the crowd, and this posture coincided with the best outcomes.
- **Activity spikes in fear, not greed.** Trade count correlates negatively with the sentiment index ($r \approx -0.24$); traders are busiest when the market is fearful.
- **Capital sizing doesn't track where the edge is.** Average position size is largest in Fear (~\$7.8K) and smallest in Extreme Greed (~\$3.1K) — inverted relative to where the historical edge actually was.
- **Extreme Greed offers the best risk-adjusted return.** Extreme Greed shows the best combination of high average return and low volatility; Fear and Greed themselves are the most volatile regimes.
- **Performance is concentrated.** 3 of 32 accounts drive ~46% of total realized PnL — aggregate sentiment patterns are influenced by a small number of large/skilled traders.

9. Strategic Implications

- Consider a sentiment-conditioned sizing rule: scale position size up in Extreme Greed and down in Fear/Extreme Fear, aligning capital deployment with the regime that historically carried the strongest risk-adjusted edge in this sample.
- Formalize the contrarian directional bias already present in trader behavior (long-in-fear, short-in-greed) into a systematic signal, and validate it with proper walk-forward backtesting before any live deployment.
- Segment traders by sentiment-sensitivity — some accounts are regime-agnostic performers while others are highly sentiment-dependent; risk limits and capital allocation could be tailored accordingly.
- Treat all findings here as exploratory hypotheses, not confirmed trading edges: the trade dataset spans roughly five months versus a multi-year sentiment index, and only 32 accounts are represented. Broader, longer-horizon data would be needed before committing capital against these patterns.

Full reproducible analysis code and cell-by-cell outputs are provided in the accompanying Jupyter notebook ([trader_sentiment_analysis.ipynb](#)).